

Social Vulnerability and Compound Heat-Flood Exposure in Houston, Miami, and Norfolk: A Census-Tract-Level Analysis

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Abstract

Urban coastal communities in the United States face simultaneous chronic exposure to extreme heat and flooding, yet these hazards are typically studied and managed in isolation. This study examines whether co-located heat and flood exposure, termed compound exposure potential, is disproportionately concentrated in socially vulnerable census tracts across three U.S. coastal cities with contrasting flood mechanisms: Houston (Texas), Miami (Florida), and the Norfolk cluster (Virginia). Daytime land surface temperature composites from Landsat 8 and Landsat 9 imagery (2019 to 2023) and National Flood Insurance Program paid claim density are combined to classify 1,383 census tracts into four compound exposure classes. Social vulnerability is measured using the Centers for Disease Control and Prevention Social Vulnerability Index 2022 and tested as a predictor of compound exposure using ordinary least squares, logistic, quantile, and diagnostic-driven spatial regression. Compound-high tracts show significantly higher social vulnerability (mean 0.67 versus 0.51 for low-exposure tracts), lower median household income (68,037 versus 88,155 US dollars), and more pre-1980 housing. Social vulnerability significantly predicts compound exposure (odds ratio 4.1), and the association strengthens among the most exposed tracts and varies by city, strongest in Miami, significant but attenuated in Houston, where Hurricane Harvey's inundation obscures the chronic flood signal, and non-significant in Norfolk. Bivariate spatial clustering and geographically weighted regression show that Houston's modest citywide association masks locally concentrated, substantially stronger relationships in specific neighborhoods. Compound heat-flood exposure tracks social inequality additively rather than multiplicatively, with the most exposed tracts consistently the most socially vulnerable.

Introduction

Coastal urban communities in the United States are increasingly exposed to two hazards that intensify under climate change: extreme heat and flooding. Heat exposure, mediated by the urban heat island effect, imposes mortality and morbidity burdens that fall disproportionately on low income and minority populations (Chakraborty et al. 2019; Hsu et al. 2021; Benz and Burney 2021; Johnson 2022). Historical processes of racial residential segregation and disinvestment, including federally sanctioned redlining, have produced landscapes in which communities of color are systematically exposed to greater impervious surface cover, lower tree canopy, and consequently higher ambient temperatures (Mitchell and Chakraborty 2015; Hoffman et al. 2020). Flood risk, shaped by topography, drainage infrastructure, and floodplain development histories, similarly concentrates in communities with fewer resources to adapt or recover (Elliott and Pais 2006; Howell and Elliott 2019; Qiang 2019; Flores et al. 2023; Galster et al. 2024). Tract-level studies in Miami and Houston have documented disproportionate flood risk in minority and low-income neighborhoods (Chakraborty et al. 2014; Collins et al. 2019), and a direct comparison of the two cities found that the social patterning of flood risk differs by metropolitan context rather than following a single national template (Collins et al. 2018). At the national scale, flood exposure is likewise unevenly distributed across social vulnerability strata (Tate et al. 2021; Messenger et al. 2021). The concept of social vulnerability, foundational to this literature, was formalized by Cutter et al. (2003) and has since been refined through case-based and methodological critique (Cutter et al. 2000; Tate 2013; Rufat et al. 2015; Spielman et al. 2020), work that informs the interpretation of the vulnerability index used in the present study. Despite consistent evidence that each hazard is socially patterned, far less is known about how exposure is distributed when communities face both hazards at once.

The compound hazard literature has grown substantially since Zscheischler et al. (2020) formalized a typology of co-occurring environmental stressors, building on earlier work establishing that interacting extremes can generate risks exceeding the sum of their individual effects (Raymond et al. 2020). The Fifth National Climate Assessment (USGCRP 2023) identifies compound events, including concurrent heat and flood, as a growing threat to U.S. coastal communities, and notes that these risks fall most heavily on low-income communities and communities of color. Zhao et al. (2024) project that the annual number of compound flood-heatwave days will increase roughly sevenfold by 2100, with exposure rising fastest in lower-income regions. For coastal cities, co-located high heat and chronic flood exposure is plausibly more burdensome than either hazard alone: floodplain housing tends to be older and less energy-efficient, which limits passive cooling; flood debris and blocked drainage can trap heat; and flood damage can disrupt access to cooling centers. In Houston specifically, pluvial flooding follows complex, multi-causal pathways linked to impervious development and drainage capacity (Feng et al. 2023), conditions that plausibly interact with the urban heat island in ways that remain unexamined at the census-tract scale. Yet no study has systematically mapped this co-located exposure at the census-tract scale across multiple coastal cities or tested whether it is socially patterned.

This study addresses that gap by drawing on three research domains. Compound hazard theory frames co-located heat and flood exposure as compound exposure potential, distinct from a temporally coincident compound event (Zscheischler et al. 2020). Social vulnerability and environmental justice scholarship documents the systematic concentration of environmental burdens in marginalized communities through historical processes of discriminatory planning and disinvestment (Mohai et al. 2009; Flanagan et al. 2011; Bullard 2018; Pulido 2000). The double exposure framework (O'Brien and Leichenko 2000) situates hazard exposure within broader socioeconomic vulnerability. Together these traditions predict that compound exposure potential, like single-hazard exposure, will track patterns of social inequality rather than fall randomly across the urban landscape.

Three research questions follow. First, is compound heat-flood exposure concentrated in socially vulnerable census tracts? Second, do the two hazards combine additively, or does their co-occurrence amplify the vulnerability association beyond the sum of each hazard alone? Third, is the vulnerability-exposure relationship consistent across cities with different flood mechanisms and, if not, is that inconsistency uniform within a city or spatially concentrated? To answer these, 1,383 census tracts across Houston, Miami, and the Norfolk cluster are analyzed in five steps: derivation of tract level heat and flood exposure metrics from satellite imagery and flood insurance records; classification of tracts into compound exposure categories; statistical testing of the social vulnerability-exposure association; city-specific and diagnostic-driven spatial modeling; and local spatial analysis of within-city variation in the vulnerability-exposure relationship. Throughout, compound exposure potential (the spatial co-location of chronic heat and flood risk) is distinguished from compound events (which require temporal coincidence), a distinction that is essential for interpreting the findings accurately.

Study Area

Three coastal study areas were selected to maximize variation in flood mechanism, climate regime, and demographic composition while remaining within the U.S. federal open-data ecosystem (Fig. 1). Harris County, Texas (Houston) experiences predominantly pluvial and riverine flooding driven by intense rainfall over an extensively impervious, low-gradient landscape, combined with a humid subtropical heat regime. Miami-Dade County, Florida (Miami) faces both coastal storm surge and rainfall-driven pluvial flooding in a subtropical climate with high baseline temperatures. The Norfolk cluster, comprising the independent cities of Norfolk, Virginia Beach, Chesapeake, and Portsmouth in southeastern Virginia, experiences chronic coastal inundation accelerated by some of the fastest rates of relative sea-level rise on the U.S. Atlantic coast, where land subsidence compounds global sea-level rise. Historical redlining and racially discriminatory planning in the Hampton Roads region concentrated minority populations in low-lying, flood-prone neighborhoods (Michel et al. 2024).

Figure 1 Study area. a. Contiguous United States with study area locations, b. Harris County, Texas (Houston), c. Miami-Dade County, Florida (Miami), d. Norfolk cluster (Norfolk, Virginia Beach, Chesapeake, and Portsmouth), Virginia

All analyses use the 2020 U.S. Census Bureau census tract as the unit of analysis, with county or county-equivalent jurisdictions defining the study boundary. The three study areas collectively contain 2,107 census tracts, of which 1,383 were retained after excluding tracts that lacked valid satellite-derived heat observations, Social Vulnerability Index scores, or housing data needed to compute flood exposure. Site selection was purposive rather than random; the findings therefore support analytical generalization to coastal cities sharing comparable hazard mechanisms, not statistical generalization to all U.S. coastal cities.

Heat exposure

Images were limited to the summer months (June through September) across 2019 to 2023 and screened for cloud contamination using the QA_PIXEL quality band, masking dilated cloud, cloud, cloud shadow, and cirrus pixels. Open water was removed using the Joint Research Centre Global Surface Water dataset (Pekel et al. 2016), which prevents coastal tracts from being biased toward artificially low temperatures. A median composite was computed across all valid summer pixels within each city, and the mean land surface temperature was extracted for each 2020 census tract using the reduceRegions function in Google Earth Engine. Scaled band values were converted to degrees Celsius using the scale factor and offset specified by USGS (Eq. 1):

$$LST (^{\circ} C) = ST_B10 \times 0.00341802 + 149.0 - 273.15$$

1

where LST is land surface temperature and ST_B10 is the scaled Landsat Collection 2 Level-2 surface temperature band. The resulting measure captures daytime surface temperature near 10:30 local solar time and should not be equated with air temperature or nighttime heat exposure; it is therefore referred to as daytime surface heat throughout.

Flood exposure

Claims were filtered to the six study jurisdictions using the dataset's county code field and aggregated to 2020 census tracts using the native census tract identifier. Claims recorded before 2020 carry 2010-era tract codes; these were harmonized to 2020 boundaries using the NHGIS 2010-to-2020 census tract crosswalk (Manson et al. 2023), weighted by housing-unit allocation. Claim density was computed as total paid claims divided by tract housing units (ACS 2018 to 2022, variable B25001_001) and expressed as claims per 100 housing units.

In Houston, Hurricane Harvey (2017) accounted for 28.8 percent of all Harris County claims and 61.4 percent of total paid claim dollars. Harvey's near-county-wide spatial extent effectively randomized flood damage across socioeconomic strata: the Spearman rank correlation between Harvey-specific claim density and social vulnerability was 0.07 ($p = 0.09$), compared with 0.17 ($p < 0.001$) for non-Harvey claims. Houston's flood measure was therefore computed from non-Harvey claims only, which preserves the chronic flood exposure signal. This decision is treated as a finding in its own right, because it shows how a single catastrophic but spatially indiscriminate event can mask the chronic relationship between vulnerability and exposure. A quartile-based version of the compound exposure classification was retained as a robustness check; results were consistent in direction and significance with the primary median-split classification and are reported in Online Resource. FEMA Special Flood Hazard Area coverage was considered as an alternative flood measure but was not incorporated into the final analysis. NFIP claims capture only insured flood losses; because lower-income households and communities of color are less likely to hold flood insurance and less likely to receive full claim payments (Highfield et al. 2013; Gotham 2015; CBO 2024), this measure likely understates flood exposure in the most vulnerable tracts, a bias that is conservative with respect to the central hypothesis.

Data and Methods

Data sources

All data are freely available from federal open-data repositories (Table 1). Heat exposure was derived from Landsat 8 and Landsat 9 Collection 2 Level-2 surface temperature products (USGS 2024), accessed through Google Earth Engine (Gorelick et al. 2017). Because Landsat 9 became operational in February 2022, it contributes only to the 2022 and 2023 portions of the composite. Flood exposure was based on the Federal Emergency Management Agency (FEMA) National Flood Insurance Program (NFIP) Redacted Claims v2 dataset, accessed through the OpenFEMA application programming interface (FEMA 2026). Social vulnerability was measured with the Social Vulnerability Index 2022 (SVI) published by the Centers for Disease Control and Prevention and the Agency for Toxic Substances and Disease Registry (Flanagan et al. 2011; CDC 2024), which is built on 2018 to 2022 American

Community Survey (ACS) five-year estimates and tabulated to 2020 census tracts. The primary vulnerability measure is the overall SVI percentile rank, here denoted RPL_THEMES, which ranges from 0 (least vulnerable) to 1 (most vulnerable) and combines four themes: socioeconomic status, household characteristics, racial and ethnic minority status, and housing type and transportation.

Table 1
Data sources, derived variables, spatial resolution, and temporal coverage

Dataset	Variable	Source	Resolution	Period
Landsat 8/9 C2 L2 (ST_B10)	Daytime summer land surface temperature, water-masked	USGS via Google Earth Engine	30 m raster	Jun-Sep 2019–2023
NFIP Redacted Claims v2	Paid claim density per 100 housing units	OpenFEMA API	Census tract	1978–2024
SVI 2022	Overall percentile rank and theme scores	CDC/ATSDR	Census tract (2020)	2018–2022 ACS
ACS 2018–2022 (5-year)	Income, housing age, renter share	U.S. Census Bureau	Census tract (2020)	2018–2022
TIGER/Line 2020	Census tract boundaries	U.S. Census Bureau	Polygon	2020

Source values of -1, which indicate missing data, were recoded as missing. Three covariates were drawn from ACS 2018 to 2022 five-year estimates: the percentage of housing units built before 1980 (Pct_Pre1980; variable series B25034), the percentage of renter-occupied units (Pct_Renter; B25003), and median household income (MedIncome, in US dollars; B19013_001). The overall analytical workflow, from data source to spatial analysis, is summarized in Fig. 2.

Compound exposure classification

Within-city percentile ranks were computed separately for each study area for heat (LST_pct, where 0 is the coolest and 1 the hottest tract in that city) and flood exposure (Flood_pct01, where 0 is the lowest and 1 the highest claim density in that city), both scaled to the interval from 0 to 1. A within-city median split on each variable produced a four-class typology: Compound-high (heat and flood both at or above the city median), Heat-dominant (heat at or above, flood below), Flood-dominant (heat below, flood at or above), and Low-exposure (both below). Standardizing within each city prevents differences in climate and flood history between cities from driving the classification. A continuous Compound Exposure Index, ranging from 0 when both exposures are minimal to approximately 1 when both are maximal, was defined as the product of the two percentile ranks (Eq. 2):

$$Compound_index = LST_pct \times Flood_pct01$$

2

This product is high only when both exposures are simultaneously elevated, which makes it interpretable as a measure of joint exposure intensity. A quartile-based version of the classification was computed as a robustness check (Online Resource).

Statistical analysis

Differences in vulnerability across the four exposure classes were assessed with Kruskal-Wallis tests, which are appropriate for the skewed distribution of the vulnerability index, followed by Dunn post-hoc pairwise comparisons with Holm correction for multiple testing. Spearman rank correlations between the Compound Exposure Index and individual vulnerability indicators were also computed.

The regression models treated compound exposure as the outcome and social vulnerability as a predictor. This direction avoids the circularity that would arise from regressing the vulnerability composite on its own component variables. Four specifications were estimated. The baseline model (M1) was an ordinary least squares regression with heteroskedasticity-consistent (HC3) standard errors (Eq. 3):

$$Compound_index = \beta_0 + \beta_1 RPL_THEMES + \beta_2 Pct_Pre1980 + \beta_3 Pct_Renter + \beta_4 City_FE + \epsilon$$

3

where RPL_THEMES is the overall SVI percentile rank (0 = least vulnerable, 1 = most vulnerable); Pct_Pre1980 is the percentage of housing units built before 1980, used as a proxy for housing age and energy efficiency; Pct_Renter is the percentage of renter-occupied units; City_FE denotes city fixed effects (indicator variables for Miami and Norfolk, with Houston as the reference); and ϵ is the error term. The second model (M2) was a binary logistic regression in which the outcome equaled 1 for compound-high tracts and 0 otherwise, using the same predictors. The third model (M3) tested whether the two hazards combine beyond additivity by regressing an external vulnerability variable (either Pct_Pre1980 or MedIncome) on binary heat and flood indicators and their interaction; the heat and flood indicators equal 1 when a tract is at or above the within-city median on that exposure. The fourth model (M4) was a quantile regression estimated across quantiles from the 10th to the 90th percentile of the Compound Exposure Index in increments of 5 percentiles, with the 50th, 75th, and 90th percentiles highlighted in Results; this tests whether the vulnerability signal intensifies among the most exposed tracts. Bootstrap standard errors were used with 500 replications. City fixed effects were included in all pooled models, and every model was also estimated separately for each city to assess cross-city consistency.

Spatial diagnostics and local spatial analysis

Spatial autocorrelation in the residuals of the baseline model was assessed with the global Moran’s I statistic using Queen contiguity weights, computed separately for each city to avoid the disconnected spatial graph that pooling geographically separate study areas would produce. Where Moran’s I was significant, Anselin’s robust Lagrange Multiplier tests were used to select between a spatial error model (SEM) and a spatial lag model (SAR) for each city, following the decision rule that the model corresponding to the larger, significant robust statistic is retained (Anselin and Rey 2014). Each selected model was fitted by maximum likelihood and compared with ordinary least squares using the Akaike Information Criterion. The spatial error and spatial lag specifications are given by Eqs. 4 and 5, respectively:

$$Compound_index = \beta_0 + \beta_1 RPL_THEMES + \beta_2 Pct_Pre1980 + \beta_3 Pct_Renter + \beta_4 City_FE + u, u = \lambda Wu + \epsilon$$

4

where u is the spatially correlated error term and λ is the spatial autoregressive coefficient for the error process.

$$Compound_index = \rho W Compound_index + \beta_0 + \beta_1 RPL_THEMES + \beta_2 Pct_Pre1980 + \beta_3 Pct_Renter + \beta_4 City_FE + \epsilon$$

5

where ρ is the spatial autoregressive coefficient for the lagged dependent variable, and W is the row-standardized queen contiguity spatial weights matrix used throughout the spatial analysis.

Bivariate Local Moran’s I (Anselin 1995) was used to test whether the Compound Exposure Index and social vulnerability cluster spatially at the tract level, classifying each tract as High-High, Low-Low, High-Low, Low-High, or not significant based on 199 conditional permutations. Geographically weighted regression (GWR; Brunsdon et al. 1996) was estimated separately for each city to test whether the vulnerability-exposure relationship varies within, and not only between, cities. An adaptive Gaussian kernel bandwidth was selected by cross-validation for each city, and local coefficients, standard errors, and approximate t-values (|t| > 1.96 treated as locally significant) were retained for every tract. All spatial analyses used the spdep, spatialreg, and spgwr packages (Bivand et al. 2013). All statistical analysis was conducted in R version 4.5, and maps were produced in ArcGIS Pro.

Results

Spatial distribution of compound exposure

The classification produced broadly balanced classes across the three cities (Fig. 3; Table 2). Compound-high tracts number 320 in total, representing 23.1 percent of the 1,383 analyzed tracts. In Houston, they concentrate in the urban core and inner-ring neighborhoods immediately south and east of downtown, where dense impervious cover drives high surface temperatures and where chronic, non-Harvey flood exposure is elevated.

Table 2
Descriptive statistics by compound exposure class (n = 1,383 tracts). Values are mean (SD). *** p < 0.001 in Kruskal-Wallis tests; asterisks in cells denote significance in Dunn post-hoc comparisons with the compound-high class

Variable	Compound-high (n = 320)	Heat-dominant (n = 371)	Flood-dominant (n = 408)	Low-exposure (n = 284)	Kruskal-Wallis chi-squared
Social vulnerability score	0.67 (0.28)	0.61 (0.28)	0.50 (0.28)	0.51 (0.29)	83.2 ***
Median income (US dollars)	68,037 (34,092)	72,223 (32,631)	89,261 (43,461)	88,155 (37,445)	95.9 ***
Pre-1980 housing (percent)	65.1 (23.9)	52.3 (30.3)	60.8 (26.2)	43.7 (31.8)	83.5 ***
Renter-occupied (percent)	46.9 (24.5)	51.4 (26.9)	34.0 (21.7)	39.0 (25.4)	100.8 ***
Minority population (percent)	77.4 (23.9)	76.7 (22.2)	63.8 (26.6)	69.0 (24.2)	81.9 ***
Land surface temperature (°C)	43.5 (4.4)	43.3 (4.5)	37.5 (3.1)	38.3 (3.3)	491.2 ***
Flood claim density (per 100 units)	13.40 (17.0)	0.79 (0.84)	14.93 (17.4)	0.91 (0.89)	965.2 ***

In Miami, they cluster in the densely built northeastern part of Miami-Dade County near Biscayne Bay; the largely uninhabited southern portion of the county, covering the Everglades and southern wetlands, is classified as flood-dominant because claim density is high while surface temperature is low. In the Norfolk cluster, compound-high tracts concentrate in central Norfolk and adjacent older neighborhoods, where the urban heat island coincides with chronic tidal flooding.

Figure 3 Compound exposure classification by census tract. a. Houston, b. Miami, c. Norfolk cluster

Social vulnerability across exposure classes

Compound-high tracts are significantly more socially vulnerable than every other exposure class on all tested indicators (Table 2; Fig. 4). Their mean vulnerability score is 0.67, compared with 0.61 for heat-dominant, 0.50 for flood-dominant, and 0.51 for low-exposure tracts (Kruskal-Wallis chi-squared =

83.2, $p < 0.001$). Dunn post-hoc tests confirm that compound-high tracts differ significantly from flood-dominant tracts ($Z = 7.96$, adjusted $p < 0.001$) and low-exposure tracts ($Z = 6.84$, adjusted $p < 0.001$), and more modestly from heat-dominant tracts ($Z = 2.73$, adjusted $p = 0.013$). Median household income in compound-high tracts, at 68,037 US dollars, is roughly 20,000 dollars lower than in flood-dominant (89,261) and low-exposure (88,155) tracts. Pre-1980 housing is most prevalent in compound-high tracts (mean 65.1 percent), consistent with the co-location of aging housing and chronic hazard exposure. Spearman correlations between the Compound Exposure Index and vulnerability indicators are all positive and significant: social vulnerability (0.22), pre-1980 housing (0.24), and minority population (0.18), all $p < 0.001$.

Figure 4 Vulnerability indicators by compound exposure class for 1,383 tracts, a. Social vulnerability score, b. Median household income, c. Pre-1980 housing stock (percent)

Regression results

The baseline model confirms that social vulnerability is a positive and significant predictor of compound exposure after controlling for housing age, renter share, and city (coefficient 0.135, HC3 standard error 0.023, $p < 0.001$; R-squared 0.094). Variance inflation factors are all below 1.7, indicating no substantive multicollinearity. The modest R-squared indicates that social vulnerability is a real but partial determinant of compound exposure, which is expected given that physical exposure also depends on topography, drainage, and development history. The logistic model yields an odds ratio of 4.1 for social vulnerability ($p < 0.001$): a tract at the top of the vulnerability distribution has roughly four times the odds of being compound-high as a tract at the bottom.

The interaction model does not support super-additive compounding. The heat-by-flood interaction term is negative and non-significant for both pre-1980 housing (-3.3 percentage points, $p = 0.095$) and median income ($-\$3,687$, $p = 0.28$). Compound-high tracts are therefore more vulnerable because they carry both hazard burdens at once, not because the two hazards multiply one another's effect.

The quantile regression shows that the vulnerability signal strengthens toward the upper end of the exposure distribution (Fig. 5). The social vulnerability coefficient rises from 0.147 at the median to 0.223 at the 75th percentile and remains elevated at 0.203 at the 90th percentile, with all three estimates significant. The tracts most intensely exposed to co-located heat and flood are thus also the most socially vulnerable, a pattern consistent with the idea of a resilience desert in which the communities least able to cope with compound exposure are precisely those most exposed to it.

Figure 5 Quantile regression coefficients for social vulnerability predicting the Compound Exposure Index (n = 1,383)

Spatial model selection and cross-city results

Moran's I on the baseline residuals is significant in all three cities (Houston 0.245, Miami 0.109, Norfolk 0.270; all $p < 0.001$). Robust Lagrange Multiplier diagnostics favored a spatial error process in Houston (robust LM-error statistic 3.42 versus 1.83 for the lag alternative, $p = 0.064$) and a spatial lag process in Miami (3.42 versus 0.66, $p = 0.064$), though neither difference reached conventional significance; the corresponding spatial models nonetheless produced substantially lower Akaike Information Criterion values than ordinary least squares in both cities (Table 3), supporting their use. In Norfolk, the robust statistics clearly favored a spatial lag process (26.29 versus 14.26, both significant at $p < 0.001$). The diagnostic-driven spatial models fit better than ordinary least squares in every city, with lower Akaike Information Criterion values throughout (Table 3).

Table 3 Per-city regression results: social vulnerability predicting Compound Exposure Index.							
City	n	Model	OLS coef.	Spatial coef.	Spatial param.	AIC (OLS)	AIC (spatial)
Houston	647	SEM	0.079 ($p = 0.018$)	0.092 ($p = 0.014$)	$\lambda = 0.421$	-242.5	-313.1
Miami	495	SAR	0.204 ($p < 0.001$)	0.168 ($p < 0.001$)	$\rho = 0.225$	-195.3	-207.4
Norfolk cluster	241	SAR	0.090 ($p = 0.116$, ns)	0.040 ($p = 0.417$, ns)	$\rho = 0.535$	-96.5	-139.8

Houston's spatial error model shows a significant vulnerability-exposure association (coefficient 0.092, $p = 0.014$), similar in magnitude to its ordinary least squares estimate (0.079). Miami's spatial lag model shows the strongest association among the three cities (coefficient 0.168, $p < 0.001$), somewhat attenuated from its ordinary least squares estimate (0.204) once spatial spillover in compound exposure is accounted for. The Norfolk cluster shows no significant association in its spatial lag model (coefficient 0.040, $p = 0.417$), consistent with its non-significant ordinary least squares result. Norfolk's spatial lag parameter is the largest of the three cities ($\rho = 0.535$), indicating substantial spillover of compound exposure between adjacent tracts.

Bivariate spatial clustering

Bivariate Local Moran's I identified spatially significant clustering between the Compound Exposure Index and social vulnerability in 26 to 31 percent of tracts across the three cities (Table 4; Fig. 6).

Table 4 Bivariate Local Moran's I cluster percentages and geographically weighted regression (GWR) local coefficient summary by city					
City	% High-High (LISA)	%Low-Low (LISA)	% Any significant (LISA)	GWR coef. range	% locally significant (GWR)
Houston	11.0	10.8	30.9	-0.240 to 0.929	17.8
Miami	8.7	9.3	26.1	-0.153 to 0.479	30.9
Norfolk cluster	7.1	14.5	28.2	-0.310 to 0.261	12.9

Houston showed the highest proportion of significant High-High clusters (11.0 percent of tracts), indicating localized pockets where high compound exposure is spatially surrounded by high social vulnerability, despite its comparatively weaker citywide regression coefficient.

Miami showed a slightly lower High-High proportion (8.7 percent) despite its stronger citywide association, while Norfolk showed the lowest High-High proportion (7.1 percent) and was instead dominated by Low-Low clustering (14.5 percent of tracts), consistent with its non-significant regression result. For Miami, the city with the strongest citywide association, Fig. 7 presents a bivariate choropleth of the Compound Exposure Index and social vulnerability terciles, illustrating that the highest values of both variables co-occur most consistently in the densely built northeastern part of the county.

Figure 7 Bivariate choropleth of Compound Exposure Index and social vulnerability terciles for Miami (Miami-Dade County)

Local spatial variation in the vulnerability-exposure relationship

Geographically weighted regression reveals substantial local variation in the social vulnerability-exposure relationship within each city, extending beyond what the citywide coefficients in Table 3 convey (Table 4; Fig. 8). In Houston, local coefficients range from -0.240 to 0.929 , far exceeding the citywide estimate of 0.079 to 0.092 ; 17.8 percent of tracts show a statistically significant local relationship, concentrated in two dense clusters in the north-central and southern parts of the county. These clusters spatially coincide with the High-High bivariate LISA clusters described above, indicating that Houston's modest citywide association reflects an average of very strong local relationships in specific neighborhoods and near-zero relationships elsewhere, rather than a uniformly weak effect. Miami shows a more spatially uniform relationship, with local coefficients ranging from -0.153 to 0.479 and 30.9 percent of tracts locally significant, consistent with its strong and pervasive citywide association. The Norfolk cluster shows local coefficients ranging from -0.310 to 0.261 , with 12.9 percent of tracts locally significant; unlike Houston and Miami, a substantial minority of Norfolk tracts (34.0 percent) show a negative local coefficient, concentrated in the northwestern part of the cluster, indicating that vulnerability and compound exposure move in opposite directions in some neighborhoods even as they align in others.

Discussion

The finding that compound-high tracts are significantly more vulnerable, lower-income, and older in housing stock than any single-hazard class supports the central hypothesis that environmental burdens are not randomly distributed but follow patterns of historical disinvestment and marginalization (Mohai et al. 2009; Bullard 2018; Pulido 2000). The logistic odds ratio of 4.1 is a substantively meaningful disparity, not merely a statistically detectable one. The over-representation of pre-1980 housing in compound-high tracts is consistent with a well-documented mechanism: discriminatory lending and zoning historically confined minority and low-income households to older, lower-quality housing in flood-prone and heat-exposed areas (Mitchell and Chakraborty 2015; Hoffman et al. 2020; Flores et al. 2023). Older housing is also typically less energy-efficient and less likely to have reliable air conditioning, which amplifies heat vulnerability in the same neighborhoods that carry elevated flood exposure.

The interaction results temper the strongest version of the compound-burden argument. Because the heat-by-flood interaction is not significant, compound-high tracts appear to accumulate two burdens additively rather than experience a multiplicative amplification. This distinction matters for policy. An additive structure implies that programs targeting either hazard alone would each deliver partial relief to compound-high communities, while integrated programs would deliver the full combined benefit. The absence of a multiplier does not weaken the case for joint prioritization; it clarifies the mechanism through which the burden accumulates.

The rise in the vulnerability-exposure association toward the upper quantiles (Fig. 5) is among the most distinctive results of this analysis. The relationship is not uniform across the distribution of compound exposure but concentrates at its upper end, where the most exposed tracts are disproportionately the most vulnerable. A standard mean regression, by estimating a single average slope, would miss this non-linearity and understate the severity of the burden at the tail. Identifying that tail is precisely where compound-hazard policy can be most efficiently targeted.

The cross-city differences are themselves a substantive result rather than a statistical inconvenience. Miami's strong association survives the switch from ordinary least squares to a spatial lag specification, the model form favored by its own Lagrange Multiplier diagnostics and remains the largest of the three cities. Because Miami-Dade has almost no topographic relief, lower-income inner-city tracts, rather than only the lowest coastal zones, carry the highest co-located burden, producing a tight alignment between physical exposure and social vulnerability (Seeteram et al. 2023). Houston's association is smaller and is best captured by a spatial error rather than a spatial lag specification, indicating that unmeasured, spatially clustered factors affect the error structure more than compound exposure spills directly across tract boundaries. Notably, the bivariate LISA and geographically weighted regression results together show that Houston has the highest proportion of significant High-High clusters among the three cities and the widest range of local coefficients, reaching values several times larger than the citywide estimate. This indicates that Houston's compound burden is real and severe but concentrated in specific, historically underinvested neighborhoods rather than distributed evenly across the county; a policy response calibrated only to the citywide average would substantially understate the burden in these concentrated areas. Harris County's historically weak land-use regulation has permitted extensive floodplain development, producing a chronic flood geography (Berke et al. 2019) that Hurricane Harvey's near-county-wide inundation temporarily obscured but did not erase. This has a clear methodological implication: flood-equity studies in cities with a single dominant flood event should separate chronic from catastrophic exposure before drawing conclusions about chronic risk.

Norfolk shows no significant vulnerability-exposure association under either ordinary least squares or its diagnostic-favored spatial lag specification. Its large spatial lag parameter ($\rho = 0.535$), the highest among the three cities, indicates that compound exposure itself, not merely the model residuals, spills substantially across tract boundaries in Norfolk. Combined with its bivariate LISA results, which show Low-Low clustering as the dominant significant pattern rather than High-High clustering, and with geographically weighted regression results showing a substantial minority of tracts with a negative local coefficient concentrated in the northwestern part of the cluster, this suggests that Norfolk's exposure and vulnerability are each spatially smooth and, in

places, locally opposed, rather than concentrated in the kind of localized high-burden pockets seen in Houston and Miami. Redlining and discriminatory planning in Hampton Roads historically concentrated Black residents in low-lying, chronically flooded neighborhoods (Michel et al. 2024); the present analysis suggests that this concentration is expressed as a broad regional pattern of exposure and vulnerability moving together, and in some areas apart, across the landscape, rather than as sharply bounded local clusters that a tract-level regression alone would detect.

Limitations

Four limitations apply. First, the analysis captures compound exposure potential, not temporally coincident compound events. Second, Landsat surface temperature reflects daytime conditions only, with a clear-sky warm bias from cloud masking. Third, NFIP claim density covers insured losses only, likely understating exposure in the most vulnerable tracts, a conservative bias. Fourth, roughly one-third of tracts were excluded for missing data, and if these differ systematically from retained tracts, results may not generalize to them. The three-city design also supports analytical rather than statistical generalization.

Conclusion

This study provides the first multi-city, tract-level analysis of how co-located heat and flood exposure relates to social vulnerability in U.S. coastal cities, combining pooled and city-specific regression with diagnostic-driven spatial models, bivariate spatial clustering, and geographically weighted regression. Compound-high tracts, those above the within-city median on both daytime surface heat and flood claim density, are significantly more socially vulnerable, lower-income, and older in housing stock than any single-hazard class. The association is additive rather than multiplicative, is significant in two of the three cities, and is notably stronger among the most exposed tracts. Local spatial analysis shows that citywide averages can themselves be misleading: Houston's modest overall association conceals spatially concentrated, substantially stronger relationships in specific neighborhoods, a pattern invisible to standard regression alone. The Harvey dilution effect in Houston offers a further methodological caution: flood-equity research that relies on claim-density metrics in cities with a dominant single flood event should separate chronic from catastrophic exposure before drawing conclusions about chronic risk.

The policy implication is concrete. Heat action plans and flood mitigation programs in U.S. coastal cities are typically administered through separate institutional channels. The compound-high tracts identified here, especially those at the upper quantiles of the exposure distribution and within the locally concentrated clusters identified by bivariate and geographically weighted analysis, warrant joint prioritization across both program types, because they carry the full additive burden of both hazards. Future work should extend this framework to additional coastal cities, incorporate nighttime heat and modeled pluvial flooding that is not limited to insurance records, and examine how compound exposure evolves as urban heat islands intensify and sea levels rise.

Declarations

Compliance with Ethical Standards

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Conflict of Interest: The authors declare that there is no conflict of interest.

Research involving Human Participants and/or Animals: This study uses publicly available, de-identified aggregate census and administrative data and involved no human participants or animals.

Data availability

All data used in this study are freely available from the cited federal repositories, and the analysis code is available from the corresponding author on request.

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Figures

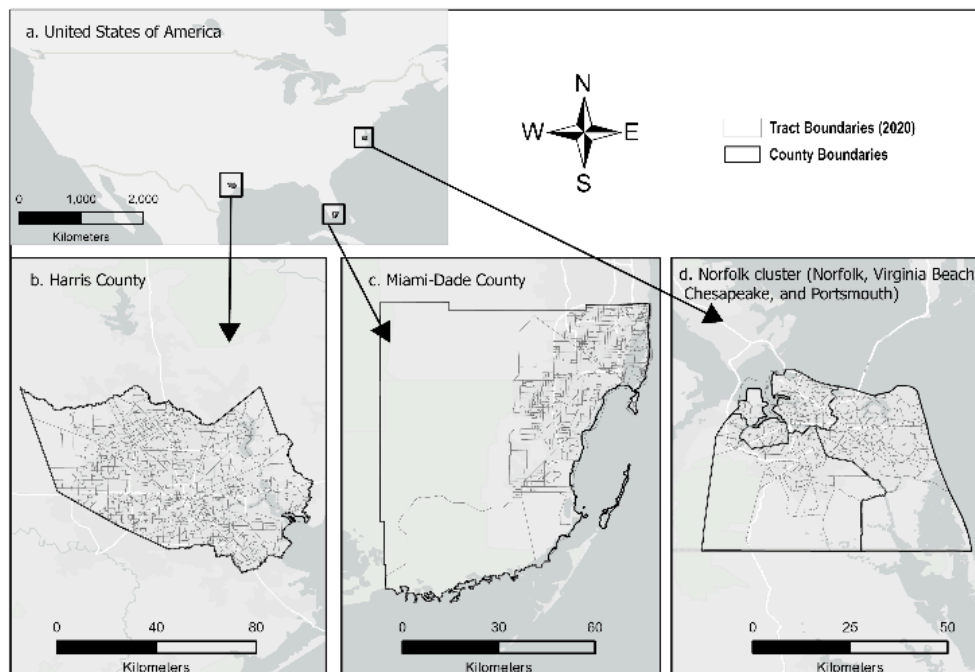


Figure 1

Study area. a. Contiguous United States with study area locations, b. Harris County, Texas (Houston), c. Miami-Dade County, Florida (Miami), d. Norfolk cluster (Norfolk, Virginia Beach, Chesapeake, and Portsmouth), Virginia

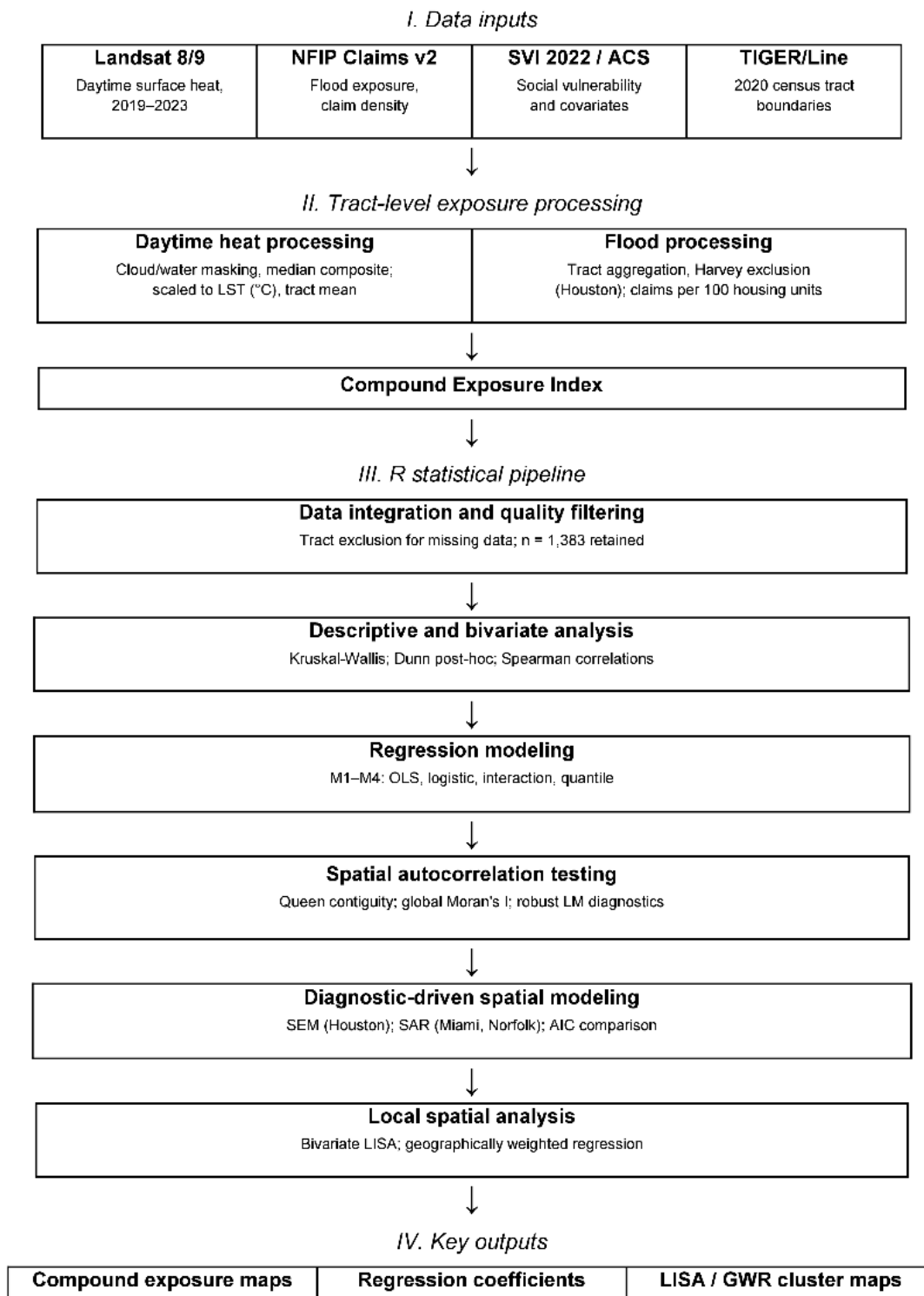


Figure 2

Methodological workflow

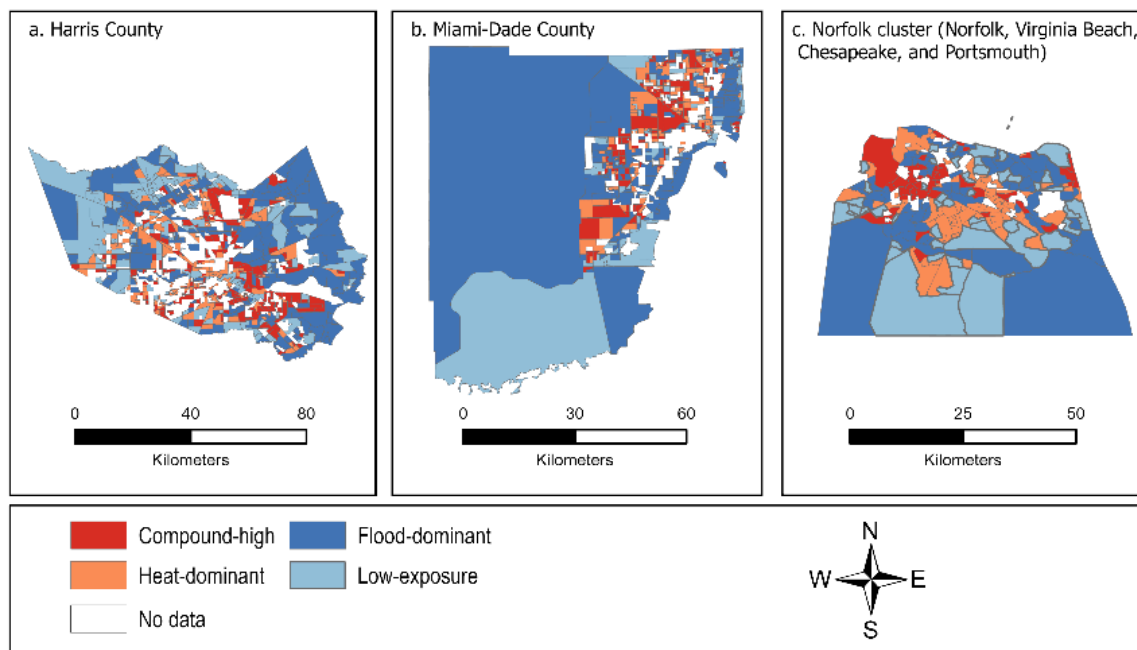


Figure 3

Compound exposure classification by census tract. a. Houston, b. Miami, c. Norfolk cluster

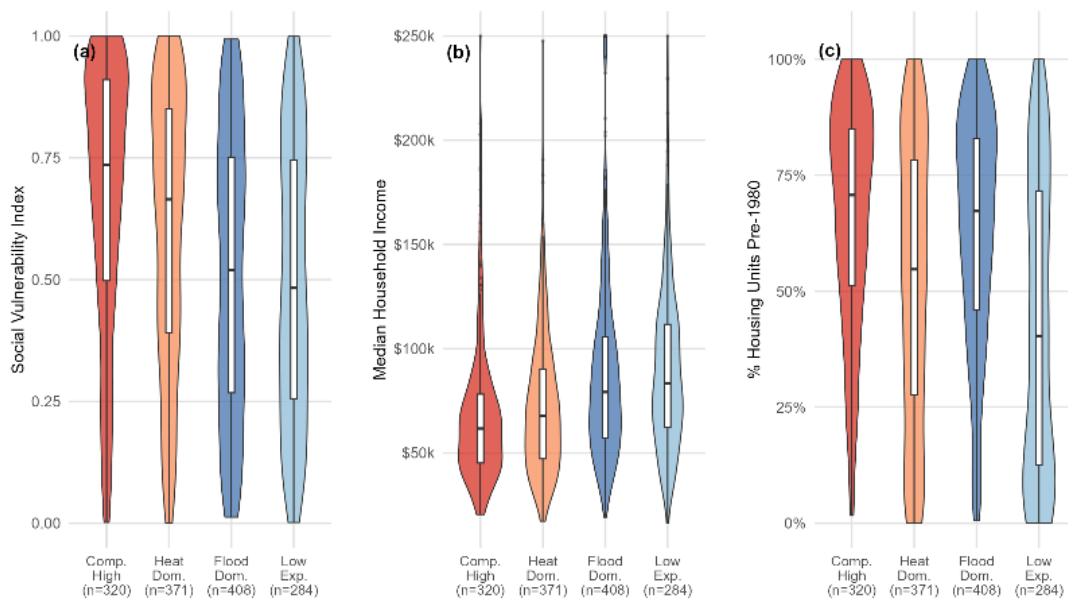


Figure 4

Vulnerability indicators by compound exposure class for 1,383 tracts, a. Social vulnerability score, b. Median household income, c. Pre-1980 housing stock (percent)

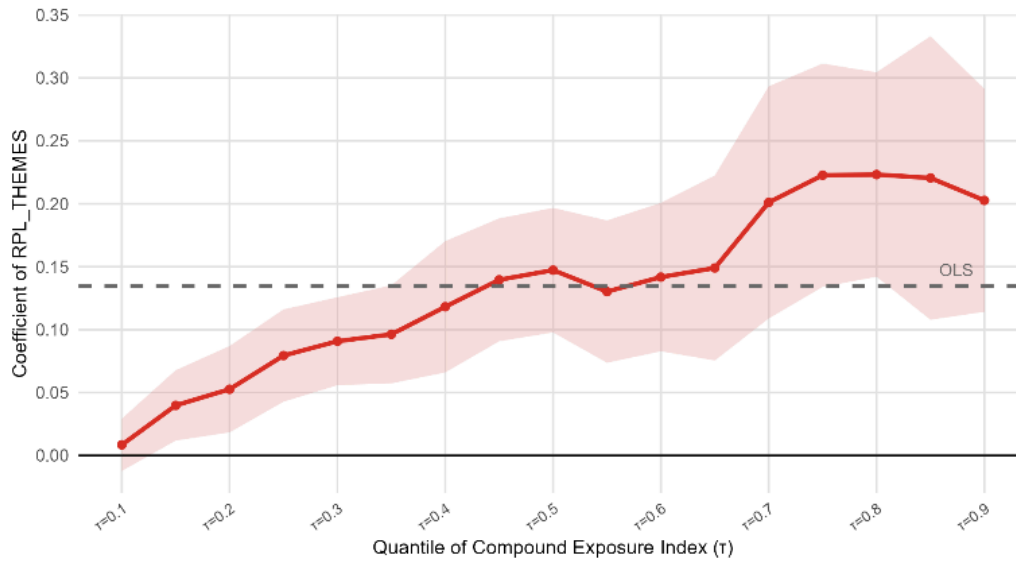


Figure 5

Quantile regression coefficients for social vulnerability predicting the Compound Exposure Index (n = 1,383)

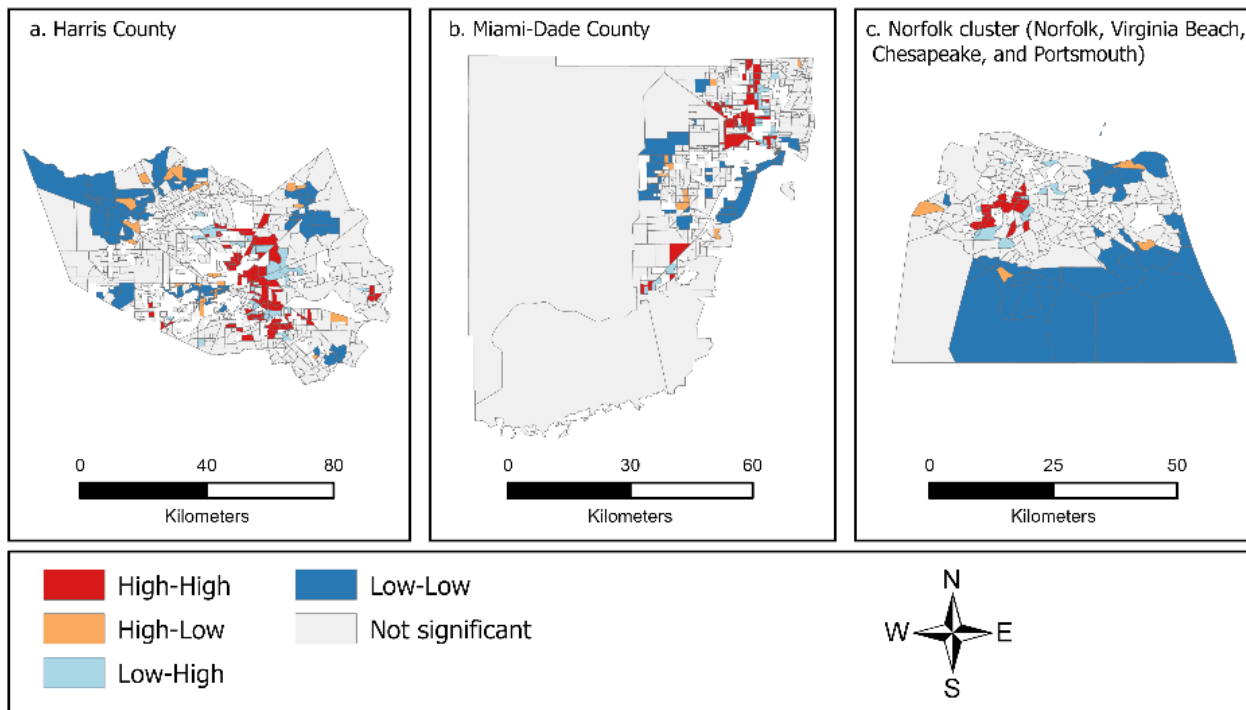


Figure 6

Bivariate Local Moran's I cluster classification between the Compound Exposure Index and social vulnerability. a. Houston, b. Miami, c. Norfolk cluster

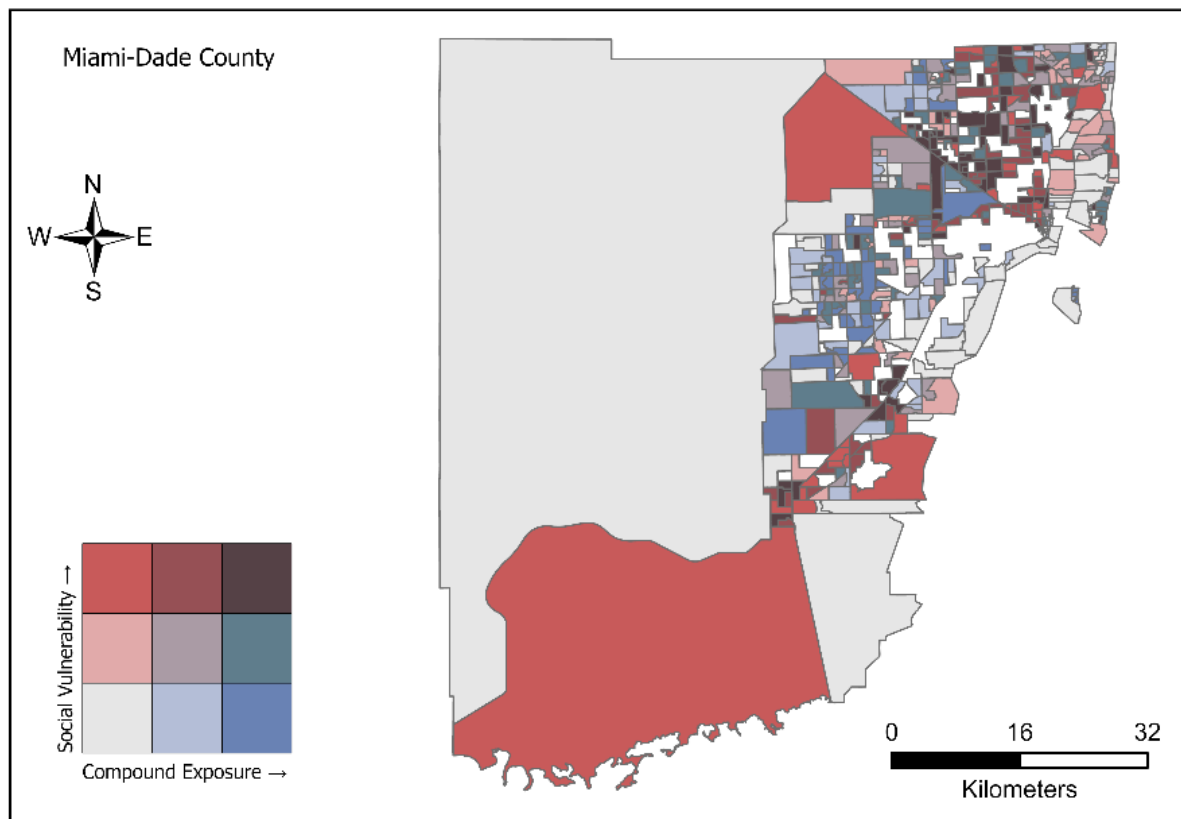


Figure 7

Bivariate choropleth of Compound Exposure Index and social vulnerability terciles for Miami (Miami-Dade County)

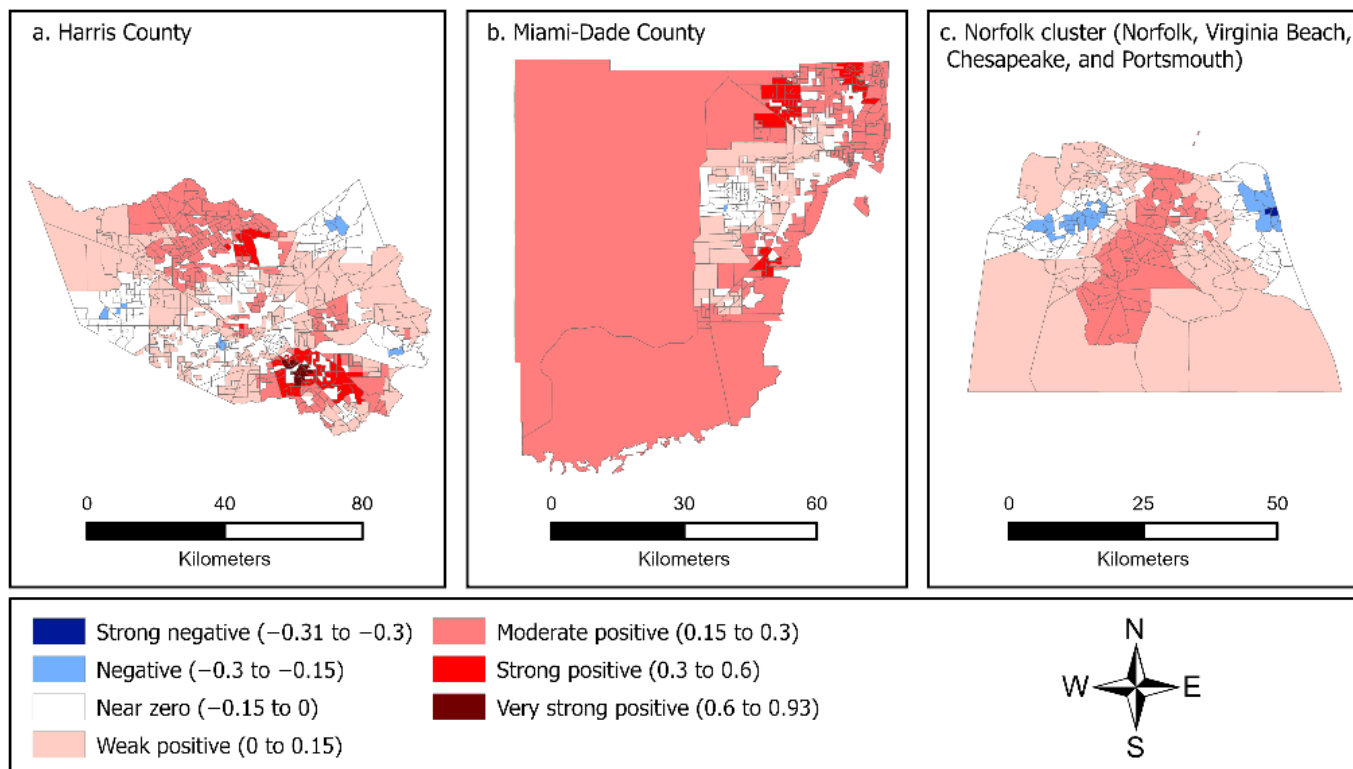


Figure 8

Local geographically weighted regression coefficients for social vulnerability predicting the Compound Exposure Index. a. Houston, b. Miami, c. Norfolk cluster

Supplementary Files

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